

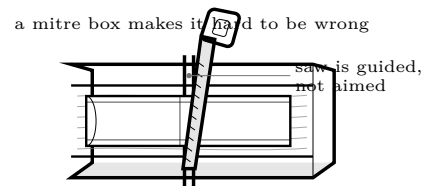
Craft 2: Cut Square

A joint can never be better than the cut that went into it

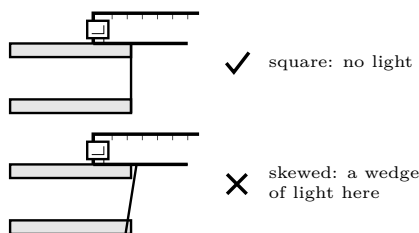
Shop skill — transferable to anything you ever build

THE STANDARD

The cut face is **perpendicular to the pipe axis** within about one degree, and the whole face is flat. Held against a try square on a flat bench, you should not see light under the blade anywhere. On a 3 in pipe, one degree of skew is about 1.3 mm of gap on the far side — enough to matter, and easy to see once you know to look.



support the pipe on *both* sides of the cut



Why square matters	A socket joint is a precision taper. A skewed pipe end bottoms on one side while leaving a gap on the other, so the cement pools where it is not wanted and the weld area is smaller than it looks from outside. The joint appears finished and is not.
Use a guide	Freehand sawing a round object square is genuinely difficult and there is no reason to try. A mitre box or a tubing cutter turns a skill problem into an equipment problem.
Tubing cutter	Best result on pipe up to about 2 in — it cuts square by construction. Tighten a quarter turn per revolution, never more, or it ploughs and leaves a lip.
Fine teeth	On plastic, a fine-toothed saw cuts cleanly and a coarse one tears. Let the saw do the work; pushing hard makes the blade wander.
Support both sides	Pipe that drops as the cut finishes tears the last of the material and takes the face out of square in the final centimetre.
Cut on the waste side	Of the line, so the line survives on the piece you are keeping. <i>see the Craft 1: Measure & Mark sheet</i>

Method

1. Mark all the way round. A cut is only as square as its mark.
2. Set the pipe in the mitre box with the mark aligned to the slot, and hold it against the fence so it cannot roll.
3. Start the cut with the saw pulled backwards a couple of times to make a kerf. Do not start on the push.
4. Saw with long strokes and light pressure, letting the guide keep the angle. Rotate the pipe a quarter turn every few strokes if the wall is thick — this keeps the cut following the mark all the way round rather than drifting on one side.
5. Support the offcut so it does not snap free.

Checking, and what to do if it is off

Test	What it tells you
Try square, light behind	The direct test. Rotate the square around the pipe to four positions — a cut can be square in one plane and skewed in the other.
Stand it on glass	A truly square end stands on a flat surface without rocking. Rock it gently; any wobble is skew.
Dry-fit and rotate	Push the pipe into its fitting and rotate it. If the gap at the shoulder changes as it turns, the end is not square.

If it is off: **re-cut, do not sand it true**. Sanding a 3 in face flat by hand takes longer than a fresh cut, removes an unpredictable amount, and usually produces a face that is flat but no longer perpendicular. Pipe is cheap. Cut again and

keep the bad piece as a test coupon for practising welds on.

Check it yourself. Hold the try square against the pipe with the blade across the cut face, and hold the pair up to a window or a lamp. You are looking for light. Then turn the square 90 degrees around the pipe and look again, and again, and again — four positions. No light in any of them means it is square.

The habit worth taking away

Every trade has a version of this: use a guide rather than your eye, and then check against a known-good reference rather than against your own judgement of how it looks. A try square is a piece of steel that is more perpendicular than you are. That is not an insult, it is the entire reason it exists.

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